

Anomaly-Based Network Intrusion Detection Using Traffic Classification

Abstract

Signature-based intrusion detection systems are effective against known attack patterns but fail to detect novel or zero-day threats. This paper presents an anomaly-based detection approach that models normal network traffic behavior and flags statistically significant deviations as potential intrusions.

1. Methodology

We collect network flow features including packet size distribution, connection duration, and protocol usage patterns, and train a classifier to distinguish normal traffic from anomalous traffic. The model is evaluated on held-out traffic captures containing both benign activity and simulated attack traffic, including port scanning and denial-of-service patterns.

2. Advantages and Limitations

The primary advantage of anomaly-based detection is the ability to catch previously unseen attack patterns that signature-based systems would miss. A limitation is a higher false-positive rate, since legitimate but unusual traffic can resemble an anomaly; careful threshold tuning and analyst review workflows are required to keep the system operationally usable.

3. Conclusion

We conclude that anomaly-based detection is a valuable complement to signature-based systems, particularly for identifying novel threats, and recommend deploying both approaches together as part of a layered network security strategy.